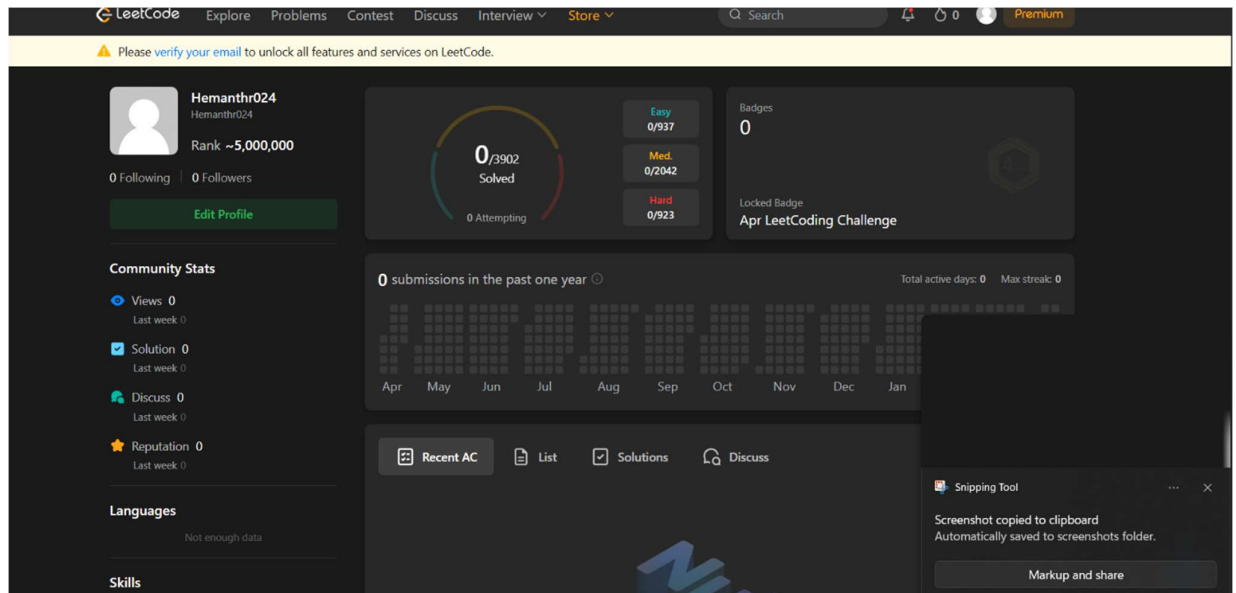


1) Create accounts on LeetCode, HackerRank, and Codeforces and solve two problems on each platform.



2) Focus on array and string problems on LeetCode and reflect on solution strategies.

3) Participate in a Codeforces contest and submit a solution analysis report.

4) Install Git and configure user credentials on your system.

```
hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (main)
$ git config user.name
hemanthkumarr2404

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (main)
$ git config user.email
hemanthkumarrbca25@rvu.edu.in
```

5) Initialize a new Git repository and commit a sample file using basic Git commands.

```
hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC
• hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC
$ mkdir "IVC_Hemanth Kumar R"

• hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC
$ cd "IVC_Hemanth Kumar R"

• hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC/IVC_Hemanth Kumar R
$ git clone https://github.com/hemanthkumarr2404/IVC_ASSINGMENT.git
Cloning into 'IVC_ASSINGMENT'...
warning: You appear to have cloned an empty repository.

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC/IVC_Hemanth Kumar R
• $ cd "IVC_ASSINGMENT"

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC/IVC_Hemanth Kumar R/IVC_ASSINGMENT (main)
• $ touch index.md

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC/IVC_Hemanth Kumar R/IVC_ASSINGMENT (main)
• $ git add .
```

6) View repository status and commit history using git status and git log.

```
manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/Chiranth_IVC (main)
• $ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1

no changes added to commit (use "git add" and/or "git commit -a")

manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/Chiranth_IVC (main)
• $ git log
commit a5fbcf81736397cc812893be8506e8f235fe4c0b (HEAD -> main, origin/main)
Author: Chiranthmbenakalbca25-tech <Chiranthmbenakalbca25@rvu.edu.in>
Date:   Wed Apr 15 20:15:21 2026 +0530

    Initial commit
```

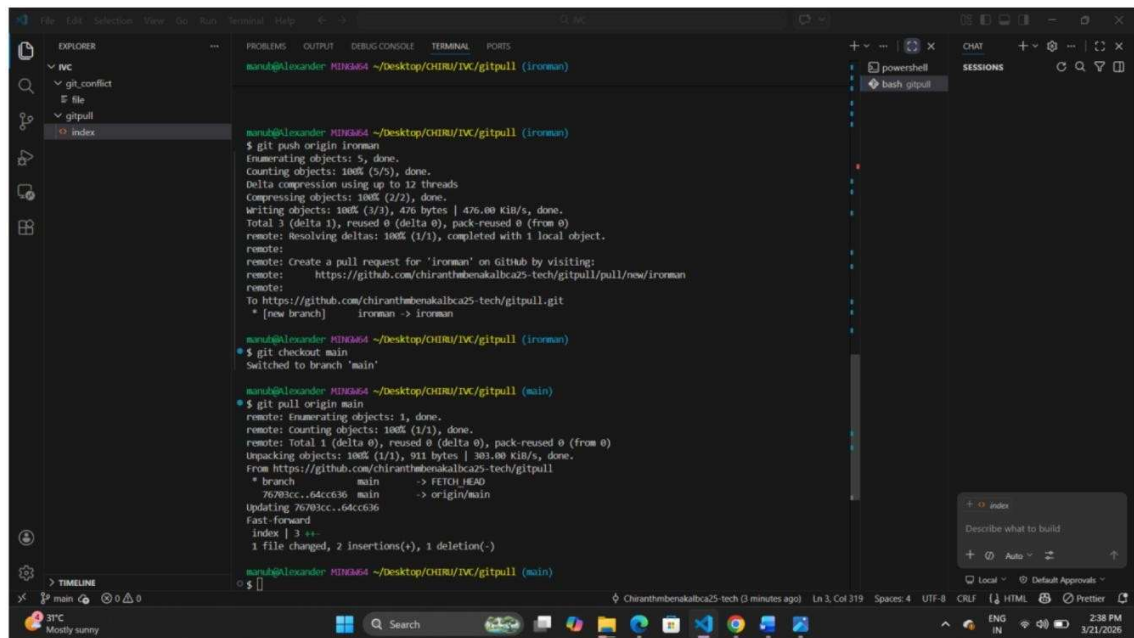
7) Create and link a local repository with GitHub and push commits using git push.

```
hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC/IVC_Hemanth Kumar R/IVC_ASSINGMENT (main)
• $ git commit -m "Updated Information"
[main (root-commit) 1675122] Updated Information
1 file changed, 2 insertions(+)
create mode 100644 index.md

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/IVC/IVC_Hemanth Kumar R/IVC_ASSINGMENT (main)
• $ git push https://github.com/hemanthkumarr2404/IVC_ASSINGMENT.git main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 24 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 271 bytes | 271.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hemanthkumarr2404/IVC_ASSINGMENT.git
* [new branch]      main -> main
```

8)

Pull updates from a remote repository and manage repository synchronization.



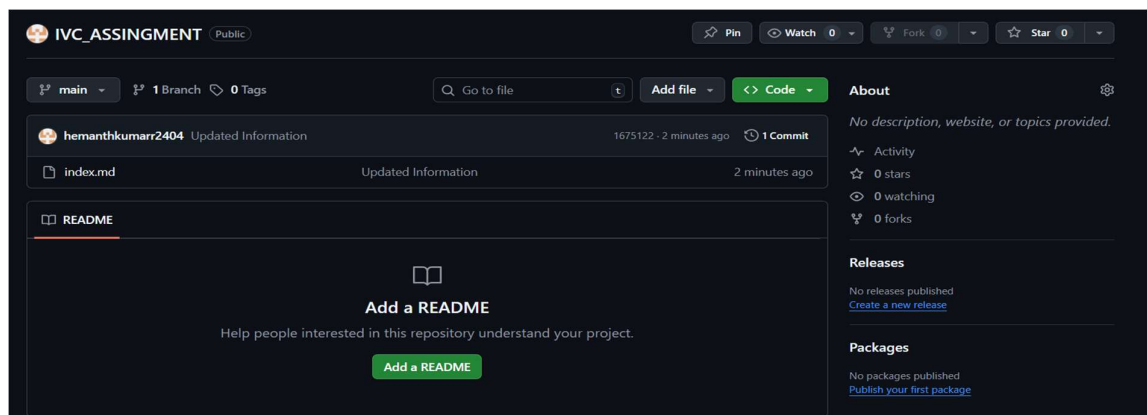
```
manuh@Alexander MINGW64 ~/Desktop/CHIRU/IVC/gitpull (ironman)
$ git push origin ironman
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 476 bytes | 476.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'ironman' on GitHub by visiting:
remote:   https://github.com/chiranthbenakalca25-tech/gitpull/pull/new/ironman
To https://github.com/chiranthbenakalca25-tech/gitpull.git
 * [new branch]   ironman -> ironman

manuh@Alexander MINGW64 ~/Desktop/CHIRU/IVC/gitpull (ironman)
$ git checkout main
Switched to branch 'main'

manuh@Alexander MINGW64 ~/Desktop/CHIRU/IVC/gitpull (main)
$ git pull origin main
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 911 bytes | 303.00 KiB/s, done.
From https://github.com/chiranthbenakalca25-tech/gitpull
 * branch      main       -> FETCH_HEAD
 7670fcc..64cc636  main    -> origin/main
Updating 7670fcc..64cc636
Fast-forward
 index | 3 ++
 1 file changed, 2 insertions(+), 1 deletion(-)

manuh@Alexander MINGW64 ~/Desktop/CHIRU/IVC/gitpull (main)
$
```

Fork and clone a GitHub repository, create a pull request with changes.



9)

The image consists of two screenshots. The top screenshot shows a VS Code terminal window with the following commands and output:

```
marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC
$ git clone https://github.com/chirubenakal/motomavericks.git
Cloning into 'motomavericks'...
Receiving objects: 100% (183/183), 46.49 MiB | 14.35 MiB/s, done.
Resolving deltas: 100% (49/49), done.

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC
$ cd Benakal

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (master)
$ git remote add upstream https://github.com/chirubenakal/motomavericks.git

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (master)
$ git fetch upstream
From https://github.com/chirubenakal/motomavericks
* [new branch]      master       -> upstream/master

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (master)
$ git checkout -b chiru
Switched to a new branch 'chiru'

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (chiru)
$ git status
On branch chiru
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (chiru)
$ git commit -m "dot at end"
[chiru f296cb9] dot at end
1 file changed, 1 insertion(+), 1 deletion(-)

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (chiru)
$ git push origin chiru
remote: Permission to Marubenakal/motomavericks.git denied to chiranthbenakal25-tech.
fatal: unable to access 'https://github.com/Marubenakal/motomavericks.git/': the requested URL returned error: 403

marud@Alexander MINKA64 ~/Desktop/CHIRU/IVC/Benakal (chiru)
```

The bottom screenshot shows a GitHub pull request page for "About program1 #1". The pull request is merged and closed. The page shows the commit history, including a commit by "hemanthkumar2404" merged into "main". The pull request is successfully merged and closed, and the branch can be safely deleted.

Create feature branches, switch between them, and merge them using Git.

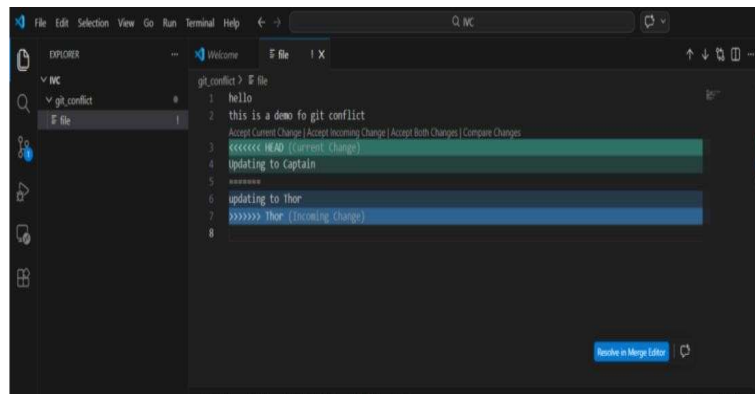
10)

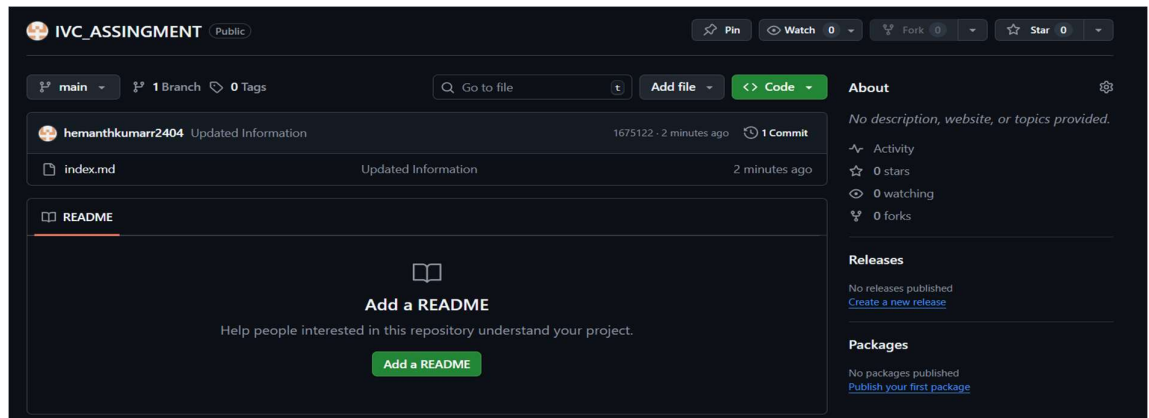
```
manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/git_conflict (Captain)
$ git push origin Captain

manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/git_conflict (Thor)
$ git checkout Captain
Switched to branch 'Captain'

manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/git_conflict (Captain)
$ git merge Thor
Auto-merging file
CONFLICT (content): Merge conflict in file
Automatic merge failed; fix conflicts and then commit the result.
```

11) Simulate and resolve a merge conflict in a collaborative project setup.





12) Use git stash to save and apply temporary changes during development.

```
manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/Chiranth_IVC (main)
$ git stash
Saved working directory and index state WIP on main: 68d206d Updating file1

manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/Chiranth_IVC (main)
$ git stash
Saved working directory and index state WIP on main: 68d206d Updating file1

manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/Chiranth_IVC (main)
$ git stash list
stash@{0}: WIP on main: 68d206d Updating file1
stash@{1}: WIP on main: 68d206d Updating file1

manub@Alexander MINGW64 ~/Desktop/CHIRU/IVC/Chiranth_IVC (main)
$ git stash apply
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1

no changes added to commit (use "git add" and/or "git commit -a")
```

13) Rebase a feature branch onto the main branch using git rebase

```
hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (main)
$ git checkout -b feature
Switched to a new branch 'feature'

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (feature)
$ git fetch origin

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (feature)
$ git add lab.c

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (feature)
$ git commit -m "update"
[feature ae3f61b] update
2 files changed, 2 insertions(+)
create mode 100644 cie.c
create mode 100644 lab.c

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (feature)
$ git rebase origin/main
Current branch feature is up to date.

hemanth@LAPTOP-4E89TMVC MINGW64 ~/OneDrive/Desktop/ivs_git/git_ivc (feature)
$
```

14) Tag specific commits using git tag and cherrypick selected commits into another branch.


```

chiranthbenakal@Alexander MINGW64 ~/projects/sample_repo (feature-branch)
$ git log --online
a1b2c3d Fix login bug
d4e5f6g Add new feature
b7c8d9e Initial commit

$ git tag -a v1.0 a1b2c3d -m -m "Login bug fix release"
Tagged commit a1b2c3d with tag 'v1.0'.

$ git push origin v1.0
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (0/0), done.
* [new tag] v1.0 -> v1.0
To https://github.com/chiranthbenakal/sample_repo.git
* [new tag] v1.0 -> v1.0
chiranthbenakal@Alexander MINGW64 ~(main)
$ git checkout main
$ git cherry-pick a1b2c3d
[main 975a4d6] Fix login bug
Date: Tue Apr 23 12:45:30 2024 +0530
1 file changed, 1 insertion(+), 1 deletion(-)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/chiranthbenakal/sample_repo.git
# d4c3f5e..975a4d6 main -> main
chiranthbenakal@Alexander MINGW64 ~/projects/sample_repo (main)
$

```

15) Set up GitHub Co-Pilot in VS Code and generate code suggestions for simple functions.

https://github.com/hemanthkumarr2404/git_ivc.git (Generated by using Co-Pilot)